Example 8.19

Let $g(x,y) = x^2 - yx + \sin(x+y)$, $x(t) = e^{3t}$, $y(t) = t^2$, $t \in \mathbb{R}$. Find $\dfrac{dg}{dt}$.

Solution

We shall follow the tree diagram to calculate.

So first we need to find $\dfrac{\partial g}{\partial x}$, $\dfrac{\partial g}{\partial y}$, $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$.

Now $\dfrac{\partial g}{\partial x} = 2x - y + \cos(x+y)$, $\dfrac{\partial g}{\partial y} = -x + \cos(x+y)$, $\dfrac{dx}{dt} = 3e^{3t}$ and $\dfrac{dy}{dt} = 2t$.

Thus

$$\begin{aligned}
\frac{dg}{dt} &= \frac{\partial g}{\partial x}\frac{dx}{dt} + \frac{\partial g}{\partial y}\frac{dy}{dt} \\
&= \big(2x - y + \cos(x+y)\big)3e^{3t} + \big(-x + \cos(x+y)\big)(2t) \\
&= (2e^{3t} - t^2 + \cos(e^{3t} + t^2))3e^{3t} + (-e^{3t} + \cos(e^{3t} + t^2))(2t) \\
&= 6e^{6t} - 3t^2e^{3t} + 3e^{3t}\cos(e^{3t} + t^2) - 2te^{3t} + 2t\cos(e^{3t} + t^2).
\end{aligned}$$

Also, some times our $W(x,y)$ will be such that $x = x(s,t)$, and $y = y(s,t)$ where $s,t \in \mathbb{R}$. Then W can be considered as a function that depends on s and t. If x, y both have partial derivatives with respect to s,t and W has partial derivatives with respect to x and y, then we can calculate the partial derivatives of W with respect to s and t using the following theorem.

Theorem 8.3

Suppose that $W(x,y)$ is a function of two variables x, y having partial derivatives $\dfrac{\partial W}{\partial x}, \dfrac{\partial W}{\partial y}$. If both variables $x = x(s,t)$ and $y = y(s,t)$, where $s,t \in \mathbb{R}$, have partial derivatives with respect to both s and t, then

$$\frac{\partial W}{\partial s} = \frac{\partial W}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial W}{\partial y}\frac{\partial y}{\partial s}, \qquad \ldots (17)$$

$$\frac{\partial W}{\partial t} = \frac{\partial W}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial W}{\partial y}\frac{\partial y}{\partial t}. \qquad \ldots (18)$$

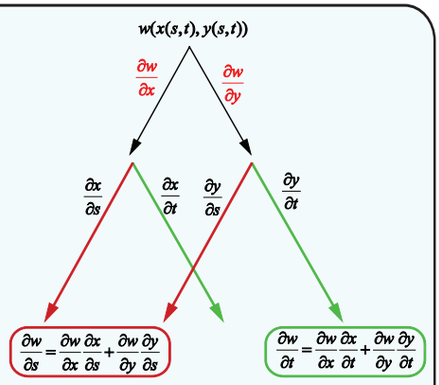


Fig. 8.15

We omit the proof. The above theorem is very useful. For instance, consider the situation in which $x = r\cos\theta$, and $y = \sin\theta, r \geq 0$ and $\theta \in \mathbb{R}$, (change from cartesian co-ordinate to polar co-ordinate system). The above theorem can be generalized for functions having n number of variables.

Let us consider an example.

Example 8.20

Let $g(x,y) = 2y + x^2, x = 2r - s, y = r^2 + 2s, r,s \in \mathbb{R}$. Find $\dfrac{\partial g}{\partial r}, \dfrac{\partial g}{\partial s}$.

Solution

Here again we shall use the tree diagram to calculate $\dfrac{\partial g}{\partial r}, \dfrac{\partial g}{\partial s}$

Hence we find $\dfrac{\partial g}{\partial x} = 2x$, $\dfrac{\partial g}{\partial y} = 2$, $\dfrac{\partial x}{\partial r} = 2$, $\dfrac{\partial x}{\partial s} = -1$, $\dfrac{\partial y}{\partial r} = 2r$, and $\dfrac{\partial y}{\partial s} = 2$.

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